Rademacher Complexity and VC Dimension

COMP 5212 – Machine Learning

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Based on Chapter 3 of *Foundations of Machine Learning*, 2nd ed.

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Where This Lecture Fits

Finite hypothesis sets union bound → Infinite hypothesis sets capacity measures → Model selection, margins, kernels, boosting

- Lecture 2 handled finite $\mathcal{H}$ using $\log|\mathcal{H}|$.
- Lecture 3 replaces cardinality by notions that remain useful when $|\mathcal{H}| = \infty$.
- The main tools are **Rademacher complexity**, the **growth function**, and **VC dimension**.

Core chapter organization follows Mohri, Rostamizadeh, and Talwalkar (2018), Chapter 3.

Learning Goals

By the end of the lecture, you should be able to:

- define empirical and expected Rademacher complexity;
- explain the symmetrization argument behind Rademacher bounds;
- connect growth functions to Rademacher complexity via Massart's lemma;
- compute or bound the VC dimension of common hypothesis classes;
- use Sauer's lemma to derive distribution-free generalization guarantees;
- interpret lower bounds in the realizable and agnostic settings.

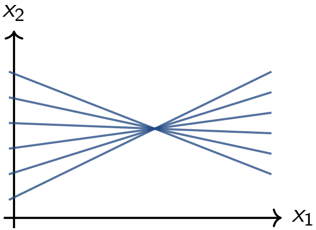
Motivation: Infinite Hypothesis Sets

- Linear classifiers contain infinitely many parameter choices (w, b).
- Intervals on $\mathbb{R}$ contain infinitely many endpoint pairs.
- Neural networks and kernel predictors form even richer classes.

The finite-class term

$$\sqrt{\frac{\log |\mathcal{H}|}{m}}$$

is meaningless when $|\mathcal{H}| = \infty$.



Uncountably many linear decision boundaries

Why the Naive Union Bound Fails

For a fixed h, concentration gives approximately

$$\mathbb{P}\left(\left|R(h) - \widehat{R}_S(h)\right| > \varepsilon\right) \leq 2e^{-2m\varepsilon^2}.$$

For finite $\mathcal{H}$, a union bound yields

$$\mathbb{P}\left(\sup_{h\in\mathcal{H}}\left|R(h) - \widehat{R}_S(h)\right| > \varepsilon\right) \leq 2\left|\mathcal{H}\right| e^{-2m\varepsilon^2}.$$

When $|\mathcal{H}| = \infty$, this gives no information. We need to measure only the distinctions the class can make on finite samples.

Three Views of Capacity

Rademacher complexity

How well can the class correlate with random signs on this sample?

Growth function

How many distinct label patterns can the class induce on m points?

VC dimension

What is the largest sample on which every binary labeling is possible?

Analytic $\longrightarrow$ combinatorial $\longrightarrow$ single-number summary

Part I: Rademacher Complexity

A data-dependent measure of the ability to fit random noise

Random Signs as a Stress Test

Fix a sample $S = (z_1, \ldots, z_m)$ and draw independent signs

$$\sigma_1, \ldots, \sigma_m \sim \mathrm{Unif}\{-1, +1\}.$$

For each $g \in \mathcal{G}$, measure the correlation

$$\frac{1}{m} \sum_{i=1}^{m} \sigma_i g(z_i).$$

A simple class cannot systematically align with independent random signs.

A rich class may select a function after seeing the signs and achieve high correlation.

Empirical Rademacher Complexity

Let $\mathcal{G}$ be a family of real-valued functions on $\mathcal{Z}$ and let $S = (z_1, \ldots, z_m)$.

$$\boxed{\widehat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i g(z_i) \right]}$$

where the σ_i are independent Rademacher variables taking values in $\{-1, +1\}$.

- The sample is fixed; the expectation is only over random signs.
- The supremum chooses the best-fitting function for each sign vector.
- The normalization by m makes the scale comparable across sample sizes.

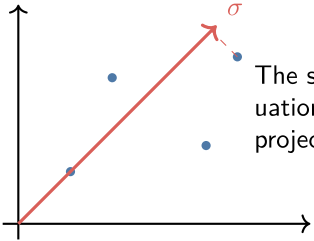
Geometry of the Definition

Associate each function with its evaluation vector

$$v_g(S) = \big(g(z_1), \ldots, g(z_m)\big) \in \mathbb{R}^m.$$

Then

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) = \frac{1}{m}\mathbb{E}_\sigma\left[\sup_{v \in A_S} \langle \sigma, v\rangle\right], \qquad A_S = \{v_g(S) : g \in \mathcal{G}\}.$$



The supremum selects the evaluation vector with the largest projection onto σ.

Toy Examples: Very Small Classes

Singleton class $\mathcal{G} = \{g_0\}$

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}_\sigma \frac{1}{m} \sum_i \sigma_i g_0(z_i) = 0.$$

There is no choice after seeing the signs.

Two constant functions $\mathcal{G} = \{-1, +1\}$

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}_\sigma \left| \frac{1}{m} \sum_i \sigma_i \right| \approx \sqrt{\frac{2}{\pi m}}.$$

The complexity decays like $m^{-1/2}$.

Rademacher complexity measures flexibility of a *class*, not variability of one fixed function.

Toy Example: A Maximally Rich Class

Suppose that on the fixed sample S, the class realizes every sign pattern:

$$\{(g(z_1), \dots, g(z_m)) : g \in \mathcal{G}\} = \{-1, +1\}^m.$$

After observing σ, choose g_σ satisfying $g_\sigma(z_i) = \sigma_i$. Then

$$\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_i \sigma_i g(z_i) = \frac{1}{m} \sum_i \sigma_i^2 = 1.$$

Therefore,

$$\boxed{\widehat{\mathfrak{R}}_S(\mathcal{G}) = 1.}$$

A class able to reproduce arbitrary random labels has maximal sample-level complexity.

Expected Rademacher Complexity

When the sample itself is random, define

$$\boxed{\mathfrak{R}_m(\mathcal{G}) = \mathbb{E}_{S\sim\mathcal{D}^m}\left[\widehat{\mathfrak{R}}_S(\mathcal{G})\right].}$$

- $\widehat{\mathfrak{R}}_S(\mathcal{G})$ is **data-dependent**: it adapts to the observed sample.
- $\mathfrak{R}_m(\mathcal{G})$ is **distribution-dependent**: it averages over samples from $\mathcal{D}$.
- A worst-case upper bound may remove dependence on both S and $\mathcal{D}$, at the cost of looseness.

Basic Properties

For a fixed sample S:

$$
\begin{aligned}
\mathcal{G} \subseteq \mathcal{F} \quad &\Longrightarrow \quad \widehat{\mathfrak{R}}_S(\mathcal{G}) \leq \widehat{\mathfrak{R}}_S(\mathcal{F}), \\
c \geq 0 \quad &\Longrightarrow \quad \widehat{\mathfrak{R}}_S(c\mathcal{G}) = c\widehat{\mathfrak{R}}_S(\mathcal{G}), \\
\mathcal{G} + a \quad &\Longrightarrow \quad \widehat{\mathfrak{R}}_S(\mathcal{G} + a) = \widehat{\mathfrak{R}}_S(\mathcal{G}), \\
\widehat{\mathfrak{R}}_S(\mathcal{G}_1 + \mathcal{G}_2) \quad &\leq \quad \widehat{\mathfrak{R}}_S(\mathcal{G}_1) + \widehat{\mathfrak{R}}_S(\mathcal{G}_2).
\end{aligned}
$$

Here a is a fixed function added to every member, and $\mathcal{G}_1 + \mathcal{G}_2 = \{g_1 + g_2 : g_j \in \mathcal{G}_j\}$.

These rules let us analyze composite function classes from simpler building blocks.

Convexification Does Not Increase Complexity

Let $\operatorname{conv}(\mathcal{G})$ denote finite convex combinations of functions in $\mathcal{G}$. For each sign vector σ,

$$\sup_{f \in \operatorname{conv}(\mathcal{G})} \sum_i \sigma_i f(z_i) = \sup_{\substack{\alpha_j \geq 0 \\ \sum_j \alpha_j = 1}} \sum_j \alpha_j \sum_i \sigma_i g_j(z_i)$$

$$= \sup_{g \in \mathcal{G}} \sum_i \sigma_i g(z_i).$$

Hence

$$\boxed{\widehat{\mathfrak{R}}_S(\operatorname{conv}(\mathcal{G})) = \widehat{\mathfrak{R}}_S(\mathcal{G}).}$$

Convex mixtures can greatly enlarge a class without increasing this complexity; this later helps analyze ensembles.

From Hypotheses to Loss Functions

Generalization concerns losses, not predictions directly. Given $\mathcal{H}$ and loss ℓ, define

$$\ell \circ \mathcal{H} = \big\{(x, y) \mapsto \ell(h(x), y) : h \in \mathcal{H}\big\}.$$

For binary $h : \mathcal{X} \to \{-1, +1\}$ with zero-one loss,

$$\mathcal{G} = \{(x, y) \mapsto \mathbf{1}\{h(x) \neq y\} : h \in \mathcal{H}\}.$$

Using $\mathbf{1}\{h(x) \neq y\} = \frac{1}{2}(1 - yh(x))$ and sign symmetry,

$$\boxed{\mathfrak{R}_m(\mathcal{G}) = \frac{1}{2}\mathfrak{R}_m(\mathcal{H}).}$$

This loss-class relation is a central specialization in Chapter 3.

Rademacher Generalization Bound

Let $\mathcal{G}$ be a family of functions $g : \mathcal{Z} \to [0,1]$. For any $\delta > 0$, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$, every $g \in \mathcal{G}$ satisfies

$$\mathbb{E}[g(z)] \leq \frac{1}{m}\sum_{i=1}^{m} g(z_i) + 2\mathfrak{R}_m(\mathcal{G}) + \sqrt{\frac{\log(1/\delta)}{2m}}.$$

A data-dependent version is

$$\mathbb{E}[g(z)] \leq \frac{1}{m}\sum_{i=1}^{m} g(z_i) + 2\widehat{\mathfrak{R}}_S(\mathcal{G}) + 3\sqrt{\frac{\log(2/\delta)}{2m}}.$$

Koltchinskii and Panchenko (2002); theorem presentation follows Chapter 3.

How to Read the Bound

population loss $\leq$ training loss + class complexity + confidence penalty.

Fit	Capacity	Confidence
$\frac{1}{m}\sum_i g(z_i)$	$2\widehat{\mathfrak{R}}_S(\mathcal{G})$	$O(\sqrt{\log(1/\delta)/m})$

- The theorem is simultaneous over all $g \in \mathcal{G}$.
- The complexity term shrinks with m for learnable classes.
- The empirical version can be smaller on “easy” samples.

Proof Roadmap

Define the worst one-sided gap

$$\Phi(S) = \sup_{g \in \mathcal{G}} \left(\mathbb{E}[g] - \widehat{\mathbb{E}}_S[g] \right).$$

The proof has three conceptual moves:

1. **Concentration:** show $\Phi(S)$ is close to $\mathbb{E}_S \Phi(S)$ by bounded differences.
2. **Ghost sample:** replace $\mathbb{E}[g]$ by an independent empirical average.
3. **Symmetrization:** random signs convert a two-sample difference into Rademacher complexity.

Uniform deviation $\Rightarrow$ randomized correlation problem

Step 1: Bounded Differences

Let S and S' differ only in the last point. Since $g \in [0,1]$,

$$\begin{aligned}
\Phi(S') - \Phi(S) &\le \sup_{g \in \mathcal{G}} \left(\widehat{\mathbb{E}}_S[g] - \widehat{\mathbb{E}}_{S'}[g] \right) \\
&= \sup_{g \in \mathcal{G}} \frac{g(z_m) - g(z'_m)}{m} \le \frac{1}{m}.
\end{aligned}$$

McDiarmid's inequality yields, with probability at least $1 - \delta/2$,

$$\Phi(S) \le \mathbb{E}_S[\Phi(S)] + \sqrt{\frac{\log(2/\delta)}{2m}}.$$

The hard part is now to bound the mean $\mathbb{E}_S[\Phi(S)]$.

Step 2: Introduce a Ghost Sample

Let $S' = (z'_1, \ldots, z'_m) \sim \mathcal{D}^m$ be independent of S. Then

$$\begin{aligned}
\mathbb{E}_S[\Phi(S)] &= \mathbb{E}_S \sup_{g \in \mathcal{G}} \left(\mathbb{E}[g] - \widehat{\mathbb{E}}_S[g] \right) \\
&= \mathbb{E}_S \sup_{g \in \mathcal{G}} \left(\mathbb{E}_{S'} \widehat{\mathbb{E}}_{S'}[g] - \widehat{\mathbb{E}}_S[g] \right) \\
&\leq \mathbb{E}_{S,S'} \sup_{g \in \mathcal{G}} \left(\widehat{\mathbb{E}}_{S'}[g] - \widehat{\mathbb{E}}_S[g] \right).
\end{aligned}$$

The inequality moves the expectation over S' outside the supremum.

We replaced an inaccessible population expectation by a comparison between two finite samples.

Step 3: Symmetrize with Random Signs

Since each pair (z_i, z_i') is exchangeable,

$$\mathbb{E}_{S,S'} \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \big(g(z_i') - g(z_i)\big)$$

$$= \mathbb{E}_{S,S',\sigma} \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i \big(g(z_i') - g(z_i)\big).$$

By subadditivity of the supremum,

$$\leq \mathbb{E}_{S',\sigma} \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_i \sigma_i g(z_i')$$

$$+ \mathbb{E}_{S,\sigma} \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_i (-\sigma_i) g(z_i).$$

Finish the Expected-Gap Bound

Because $-\sigma$ has the same distribution as σ, the two terms are equal:

$$\mathbb{E}_S[\Phi(S)] \leq 2\mathfrak{R}_m(\mathcal{G}).$$

Combining this with the bounded-differences step gives

$$\Phi(S) \leq 2\mathfrak{R}_m(\mathcal{G}) + \sqrt{\frac{\log(2/\delta)}{2m}}$$

with probability at least $1 - \delta/2$.

Symmetrization is the bridge from uniform generalization error to correlation with random noise.

From Expected to Empirical Complexity

The map $S \mapsto \widehat{\mathfrak{R}}_S(\mathcal{G})$ changes by at most $1/m$ when one point is replaced. Therefore, with probability at least $1 - \delta/2$,

$$\mathfrak{R}_m(\mathcal{G}) \leq \widehat{\mathfrak{R}}_S(\mathcal{G}) + \sqrt{\frac{\log(2/\delta)}{2m}}.$$

Combining the two events by a union bound gives

$$\Phi(S) \leq 2\widehat{\mathfrak{R}}_S(\mathcal{G}) + 3\sqrt{\frac{\log(2/\delta)}{2m}}.$$

The price for a fully observable, data-dependent bound is a slightly larger confidence constant.

Binary Classification Corollary

Let $\mathcal{H} \subseteq \{h : \mathcal{X} \to \{-1, +1\}\}$. With probability at least $1 - \delta$, every $h \in \mathcal{H}$ satisfies

$$R(h) \leq \widehat{R}_S(h) + \mathfrak{R}_m(\mathcal{H}) + \sqrt{\frac{\log(1/\delta)}{2m}}.$$

The empirical version is

$$R(h) \leq \widehat{R}_S(h) + \widehat{\mathfrak{R}}_S(\mathcal{H}) + 3\sqrt{\frac{\log(2/\delta)}{2m}}.$$

The factor $1/2$ from zero-one loss cancels the factor 2 in the general theorem.

Consequence for Empirical Risk Minimization

Define the two-sided uniform deviation

$$\Gamma_S = \sup_{h \in \mathcal{H}} \left| R(h) - \widehat{R}_S(h) \right|.$$

Let $\widehat{h} \in \arg\min_{h \in \mathcal{H}} \widehat{R}_S(h)$ and $h^* \in \arg\min_{h \in \mathcal{H}} R(h)$. Then

$$\begin{aligned} R(\widehat{h}) &\leq \widehat{R}_S(\widehat{h}) + \Gamma_S \\ &\leq \widehat{R}_S(h^*) + \Gamma_S \\ &\leq R(h^*) + 2\Gamma_S. \end{aligned}$$

Thus

$$\boxed{R(\widehat{h}) - \inf_{h \in \mathcal{H}} R(h) \leq 2\Gamma_S.}$$

Uniform convergence turns optimization of training error into a population guarantee.

A Numerical Reading of the Bound

Suppose a bounded loss class has

$$m = 1000, \qquad \widehat{\mathfrak{R}}_S(\mathcal{G}) = 0.035, \qquad \delta = 0.05,$$

and a selected predictor has empirical loss 0.08. The empirical bound gives

$$\begin{aligned} R &\leq 0.08 + 2(0.035) + 3\sqrt{\frac{\log(40)}{2000}} \\ &\approx 0.279. \end{aligned}$$

- The guarantee is rigorous but often conservative.
- Its main value is explaining scaling with sample size and class richness.
- Local or margin-based complexities can improve the value later.

Can We Compute $\widehat{\mathfrak{R}}_S(\mathcal{H})$?

The definition requires

$$\mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_i \sigma_i h(x_i) \right].$$

- For each sign vector, the inner problem resembles ERM with synthetic labels.
- Exact computation can be difficult or NP-hard for some classes.
- Monte Carlo estimates are possible when the inner optimization is tractable.
- A combinatorial upper bound can be easier: count distinct prediction vectors.

This motivates the growth function.

Checkpoint I

Discuss briefly with a neighbor:

1. Why is the supremum *inside* the expectation over signs?
2. What happens if we use only one fixed hypothesis instead of a class?
3. Why does a class realizing every labeling of m points have complexity 1?
4. Which proof step uses boundedness of the loss in $[0, 1]$?

Part II: Growth Functions and VC Dimension

Turn an infinite class into finitely many sample-level behaviors

Restriction of a Class to a Sample

For $S_X = (x_1, \ldots, x_m)$, define the set of prediction vectors

$$\mathcal{H}|_{S_X} = \{(h(x_1), \ldots, h(x_m)) : h \in \mathcal{H}\}.$$

Even if $\mathcal{H}$ is infinite, $\mathcal{H}|_{S_X}$ is finite for binary-valued hypotheses:

$$|\mathcal{H}|_{S_X}| \leq 2^m.$$

$$\boxed{\begin{array}{c}\text{Infinite parameter set}\\ \mathcal{H}\end{array}} \xrightarrow{\text{evaluate on } S_X} \boxed{\begin{array}{c}\text{Finite pattern set}\\ \mathcal{H}|_{S_X}\end{array}}$$

Growth Function

For a binary hypothesis set $\mathcal{H}$, its growth function is

$$\boxed{\Pi_{\mathcal{H}}(m) = \max_{x_1,\ldots,x_m \in \mathcal{X}} \left| \left\{ (h(x_1), \ldots, h(x_m)) : h \in \mathcal{H} \right\} \right| .}$$

- $\Pi_{\mathcal{H}}(m)$ is the maximum number of dichotomies realizable on m points.
- Always $1 \le \Pi_{\mathcal{H}}(m) \le 2^m$.
- The maximum is over point placement, not over labels.

Example: Thresholds on the Real Line

Let

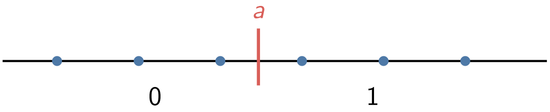
$$\mathcal{H}_{\mathrm{thr}} = \{h_a(x) = \mathbf{1}\{x \geq a\} : a \in \mathbb{R}\}.$$

On m ordered points $x_1 < \cdots < x_m$, every labeling has the form

$$00\cdots0011\cdots11.$$

There are $m+1$ possible switch locations, so

$$\boxed{\Pi_{\mathcal{H}_{\mathrm{thr}}}(m) = m+1.}$$



Example: Intervals on the Real Line

Let $h_{a,b}(x) = \mathbf{1}\{a \leq x \leq b\}$. On ordered points, the positive labels form one contiguous block.

- One labeling is all negative.
- A nonempty positive block is determined by left and right endpoints.

Therefore,

$$\boxed{\Pi_{\mathcal{H}_{\mathrm{int}}}(m) = 1 + \frac{m(m+1)}{2}.}$$

For $m = 3$, this gives 7: every labeling except 101 is realizable.

Massart's Lemma

Let $A \subset \mathbb{R}^m$ be finite and let $r = \max_{a \in A} \|a\|_2$. Then

$$\boxed{\mathbb{E}_\sigma \left[\sup_{a \in A} \frac{1}{m} \langle \sigma, a \rangle \right] \leq \frac{r\sqrt{2 \log |A|}}{m}.}$$

- Complexity depends logarithmically on the number of candidate vectors.
- The Euclidean radius r controls their scale.
- We apply the lemma to evaluation vectors of hypotheses.

Massart (2000); proof route follows the chapter's exponential-moment argument.

Massart's Lemma: Exponential-Moment Step

For any $\lambda > 0$, Jensen's inequality and the union bound give

$$\begin{aligned}
\exp\left(\lambda \mathbb{E}_\sigma \sup_{a \in A} \langle \sigma, a \rangle\right) &\leq \mathbb{E}_\sigma \exp\left(\lambda \sup_{a \in A} \langle \sigma, a \rangle\right) \\
&\leq \sum_{a \in A} \mathbb{E}_\sigma e^{\lambda \langle \sigma, a \rangle}.
\end{aligned}$$

Independence and Hoeffding's lemma imply

$$\begin{aligned}
\mathbb{E}_\sigma e^{\lambda \langle \sigma, a \rangle} = \prod_{i=1}^{m} \mathbb{E} e^{\lambda \sigma_i a_i} &\leq \exp\left(\frac{\lambda^2 \|a\|_2^2}{2}\right) \\
&\leq \exp\left(\frac{\lambda^2 r^2}{2}\right).
\end{aligned}$$

Massart's Lemma: Optimize the Parameter

Taking logarithms gives

$$\mathbb{E}_\sigma \sup_{a \in A} \langle \sigma, a \rangle \le \frac{\log |A|}{\lambda} + \frac{\lambda r^2}{2}.$$

Choose

$$\lambda = \frac{\sqrt{2 \log |A|}}{r}$$

to balance the terms. Then

$$\mathbb{E}_\sigma \sup_{a \in A} \langle \sigma, a \rangle \le r\sqrt{2 \log |A|}.$$

Dividing by m proves the displayed form.

A square root of a logarithm emerges from optimizing an exponential-moment bound.

Growth Function Bounds Rademacher Complexity

For binary functions $h : \mathcal{X} \to \{-1, +1\}$, each evaluation vector has norm $\sqrt{m}$. On any fixed sample,

$$|\mathcal{H}|_{S_X}| \leq \Pi_{\mathcal{H}}(m).$$

Massart's lemma gives

$$\widehat{\mathfrak{R}}_S(\mathcal{H}) \leq \sqrt{\frac{2 \log |\mathcal{H}|_{S_X}|}{m}} \leq \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}}.$$

Averaging over samples preserves the bound:

$$\boxed{\mathfrak{R}_m(\mathcal{H}) \leq \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}}.}$$

Generalization Bound via Growth Function

Combining the binary Rademacher bound with the growth estimate yields: with probability at least $1-\delta$, every $h \in \mathcal{H}$ satisfies

$$\boxed{R(h) \leq \widehat{R}_S(h) + \sqrt{\frac{2\log \Pi_{\mathcal{H}}(m)}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}}.}$$

- We replaced $\log|\mathcal{H}|$ by $\log \Pi_{\mathcal{H}}(m)$.
- This remains finite for every binary class on a finite sample.
- We still need a practical way to bound $\Pi_{\mathcal{H}}(m)$ for all m.

VC Dimension

The largest sample on which a class realizes every binary labeling

Shattering and VC Dimension

A set $S_X = \{x_1, \ldots, x_m\}$ is **shattered** by $\mathcal{H}$ if

$$|\mathcal{H}|_{S_X}| = 2^m.$$

The VC dimension is

$$\boxed{\mathrm{VCdim}(\mathcal{H}) = \max\{m : \Pi_{\mathcal{H}}(m) = 2^m\}.}$$

- It is a purely combinatorial notion for binary-valued classes.
- Infinite cardinality does not imply infinite VC dimension.
- Finite VC dimension means complete freedom eventually stops.

How to Prove a VC-Dimension Claim

To show $\mathrm{VCdim}(\mathcal{H}) = d$, prove two statements:

1. **Lower bound:** exhibit one set of d points that $\mathcal{H}$ shatters.
2. **Upper bound:** prove that no set of $d + 1$ points can be shattered.

Lower bounds are constructive: find a favorable configuration.

Upper bounds are universal: rule out every configuration.

A common mistake is to show only that one particular set of $d + 1$ points is not shattered.

VC Dimension of Thresholds: 1

For $h_a(x) = \mathbf{1}\{x \geq a\}$:

- One point can receive either label by moving the threshold.
- For two ordered points $x_1 < x_2$, the labeling $(1, 0)$ is impossible.

Therefore,

$$\boxed{\mathrm{VCdim}(\mathcal{H}_{\mathrm{thr}}) = 1.}$$

impossible for a one-sided threshold

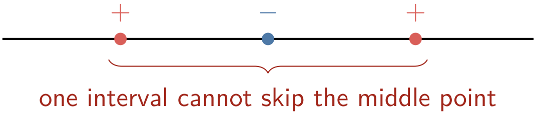
VC Dimension of Intervals: 2

Lower bound. Any two points can realize all four labelings using an interval: empty, left only, right only, or both. **Upper bound.** For three ordered points $x_1 < x_2 < x_3$, the labeling

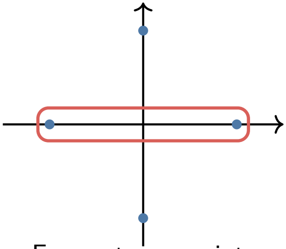
$$(+,-,+)$$

is impossible because the positive region of an interval is contiguous.

$$\boxed{\mathrm{VCdim}(\mathcal{H}_{\mathrm{int}}) = 2.}$$



Axis-Aligned Rectangles in $\mathbb{R}^2$: VC Dimension 4



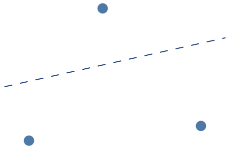
Four extreme points can be shattered.

- Lower bound: use left, right, top, and bottom extremes.
- Upper bound: among any five points, one is not an x- or y-extreme.
- Label that point negative and the four extremes positive.
- A rectangle containing all extremes must contain the fifth point.

$$\boxed{\text{VCdim(axis-aligned rectangles in } \mathbb{R}^2) = 4.}$$

Linear Separators in $\mathbb{R}^2$: VC Dimension 3

Lower bound



Three non-collinear points

Upper bound

- Four points in convex position admit an XOR labeling that no line separates.
- If one point lies inside the triangle of the other three, label the interior point positive and the outer points negative.

Every dichotomy is linearly separable.

$$\boxed{\text{VCdim(affine halfspaces in } \mathbb{R}^2) = 3.}$$

Hyperplanes in $\mathbb{R}^d$: VC Dimension $d+1$

Consider affine classifiers

$$h_{w,b}(x) = \operatorname{sign}(w^T x + b).$$

- **Lower bound:** $d+1$ affinely independent points can be shattered.
- **Upper bound:** any $d+2$ points are affinely dependent, forcing at least one nonseparable dichotomy.

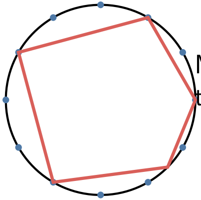
$$\boxed{\mathrm{VCdim}(\text{affine halfspaces in } \mathbb{R}^d) = d+1.}$$

The proof is geometric; parameter counting alone is not a general VC-dimension argument.

Finite and Infinite VC Dimension

Chapter 3 gives two polygon examples in the plane:

- Convex polygons with at most d sides have $\mathrm{VCdim} = 2d + 1$.
- The class of all convex polygons has infinite VC dimension.



More allowed sides permit more alternating label patterns on a circle.

Infinite VC dimension means arbitrarily large finite sets can be shattered; it is stronger than having infinitely many hypotheses.

Checkpoint II: Compute a VC Dimension

Consider unions of at most k disjoint intervals on the real line:

$$h(x) = 1 \quad \Longleftrightarrow \quad x \in \bigcup_{j=1}^{k} [a_j, b_j].$$

1. Propose a configuration giving a lower bound.
2. Identify a labeling yielding an upper bound.
3. What VC dimension do you conjecture?

Answer: $2k$. A labeling on $2k$ ordered points has at most k positive runs; the alternating labeling on $2k+1$ points has $k+1$.

Part III: Sauer's Lemma and Learning Bounds

Finite VC dimension forces polynomial growth

Sauer's Lemma

If $\mathrm{VCdim}(\mathcal{H}) = d$, then for every $m \in \mathbb{N}$,

$$\boxed{\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^{d} \binom{m}{i}.}$$

This transforms a local shattering restriction into a global bound on dichotomies.

- For $m \leq d$, the right side equals 2^m.
- For $m > d$, it is much smaller than 2^m.

Vapnik–Chervonenkis; Sauer; also independently Shelah.

Sauer's Lemma: Proof Setup

Fix $S = \{x_1, \ldots, x_m\}$ on which $\mathcal{H}$ realizes the maximum number of patterns, and let

$$G = \mathcal{H}|_S.$$

Remove the last point and write $S' = S \setminus \{x_m\}$. Define

$$\begin{aligned} G_1 &= \{g|_{S'} : g \in G\}, \\ G_2 &= \{u : u \text{ occurs in } G \text{ with both labels at } x_m\}. \end{aligned}$$

Every pattern in G_1 contributes once to G; patterns in G_2 contribute one extra copy. Hence

$$|G| = |G_1| + |G_2|.$$

Sauer's Lemma: The Key Recurrence

The two restricted families have smaller parameters:

- G_1 lives on $m-1$ points and has VC dimension at most d.
- G_2 has VC dimension at most $d-1$; otherwise a set shattered by G_2, together with x_m, would give $d+1$ shattered points.

Therefore the extremal growth quantity satisfies

$$\Pi_d(m) \leq \Pi_d(m-1) + \Pi_{d-1}(m-1).$$

The recurrence mirrors Pascal's identity for binomial coefficients.

Sauer's Lemma: Close the Induction

Apply the induction hypothesis to the recurrence:

$$
\begin{aligned}
\Pi_d(m) &\leq \sum_{i=0}^{d} \binom{m-1}{i} + \sum_{i=0}^{d-1} \binom{m-1}{i} \\
&= \sum_{i=0}^{d} \left[\binom{m-1}{i} + \binom{m-1}{i-1} \right] \\
&= \sum_{i=0}^{d} \binom{m}{i}.
\end{aligned}
$$

The boundary cases $d = 0$ and $m = 1$ are immediate.

The proof tracks which restrictions can branch into both labels at the final point.

Polynomial Growth Consequence

For $m \geq d \geq 1$,

$$\sum_{i=0}^{d} \binom{m}{i} \leq \left(\frac{em}{d}\right)^{d}.$$

Therefore,

$$\boxed{\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^{d} = O(m^{d}).}$$

Taking logarithms gives

$$\log \Pi_{\mathcal{H}}(m) \leq d \log\left(\frac{em}{d}\right).$$

This is exactly the quantity needed in the growth-function bound.

The Growth Dichotomy

A binary hypothesis class has one of two qualitatively different behaviors:

Finite VC dimension d

$\Pi_{\mathcal{H}}(m) = O(m^d)$

polynomial growth

Infinite VC dimension

$\Pi_{\mathcal{H}}(m) = 2^m$ for every m

maximal exponential growth

There is no intermediate asymptotic regime for the worst-case growth function.

VC Generalization Bound via Rademacher Complexity

Let $\mathcal{H} \subseteq \{h : \mathcal{X} \to \{-1, +1\}\}$ have VC dimension d, with $m \geq d$. Combining Sauer's lemma with the Rademacher bound gives, with probability at least $1 - \delta$, for all $h \in \mathcal{H}$,

$$\boxed{R(h) \leq \widehat{R}_S(h) + \sqrt{\frac{2d \log(em/d)}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}}.}$$

Up to logarithmic factors, the estimation term behaves like

$$\sqrt{\frac{d + \log(1/\delta)}{m}}.$$

The Standard VC Bound

A direct symmetrization and growth-function argument gives another classical form. With probability at least $1-\delta$, for all $h \in \mathcal{H}$,

$$\boxed{R(h) \le \widehat{R}_S(h) + \sqrt{\frac{8d\log(2em/d) + 8\log(4/\delta)}{m}}.}$$

It follows from a tail inequality of the form

$$\mathbb{P}\left(\sup_{h\in\mathcal{H}} \left|R(h) - \widehat{R}_S(h)\right| > \varepsilon\right) \le 4\Pi_{\mathcal{H}}(2m)e^{-m\varepsilon^2/8}.$$

Classical Vapnik–Chervonenkis bound, as compared in Chapter 3.

Comparing the Two Upper Bounds

	Rademacher route	**Classical VC route**
Main object	$\mathfrak{R}_m(\mathcal{H})$ or $\widehat{\mathfrak{R}}_S(\mathcal{H})$	$\Pi_{\mathcal{H}}(2m)$ and d
Data dependence	Possible	No
Typical scale	$\sqrt{d \log(m/d)/m}$	$\sqrt{d \log(m/d)/m}$
Strength	Flexible for real-valued and structured classes	Clean distribution-free binary result
Limitation	May be hard to compute	A single integer may be coarse

VC dimension controls worst-case capacity; empirical complexity can reveal easier data.

Sample Complexity from the VC Bound

To make the uniform gap at most ε, it suffices that

$$\sqrt{\frac{d \log(m/d) + \log(1/\delta)}{m}} \lesssim \varepsilon.$$

Solving the implicit logarithm yields

$$\boxed{m = O\left(\frac{d \log(1/\varepsilon) + \log(1/\delta)}{\varepsilon^2}\right)}$$

for agnostic uniform convergence.

- Linear dependence on capacity d up to a logarithm.
- Quadratic dependence on target accuracy $1/\varepsilon$.
- Logarithmic dependence on confidence $1/\delta$.

Worked Example: Intervals

For intervals on $\mathbb{R}$, $d = 2$. Take $m = 5000$ and $\delta = 0.05$. The Rademacher–VC deviation term is

$$\begin{aligned}
&\sqrt{\frac{2d\log(em/d)}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}} \\
&= \sqrt{\frac{4\log(2500e)}{5000}} + \sqrt{\frac{\log 20}{10000}} \\
&\approx 0.101.
\end{aligned}$$

Thus, simultaneously for all intervals,

$$R(h) \lesssim \widehat{R}_S(h) + 0.101.$$

The theorem is distribution-free and uniform, so it is often conservative compared with test-set estimates.

Worked Example: Linear Separators

Affine halfspaces in $\mathbb{R}^{20}$ have $d = 21$. With $m = 5000$ and $\delta = 0.05$,

$$\sqrt{\frac{2d \log(em/d)}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}} \approx 0.251.$$

- Capacity grows linearly with ambient dimension.
- The bound ignores margin information and observed-sample geometry.
- Margin bounds in the SVM chapter can be much sharper than raw VC dimension.

Capacity measures should match the algorithm and structure one wants to exploit.

Lower Bounds

Why the dependence on VC dimension cannot be removed

Lower Bound: Realizable Case

Let $\mathcal{H}$ have VC dimension $d > 1$. For any learning algorithm A, there exist a distribution $\mathcal{D}$ and target $f \in \mathcal{H}$ such that

$$\boxed{\mathbb{P}_{S \sim \mathcal{D}^m}\left[R_{\mathcal{D}}(h_S, f) > \frac{d-1}{32m}\right] \geq \frac{1}{100}.}$$

Consequence for constant confidence:

$$m = \Omega\left(\frac{d}{\varepsilon}\right)$$

examples are unavoidable in the realizable setting.

Ehrenfeucht, Haussler, Kearns, and Valiant (1988); Chapter 3 lower-bound theorem.

Hard Distribution Intuition

Start with d points shattered by $\mathcal{H}$:

$$X_0 = \{x_0, x_1, \ldots, x_{d-1}\}.$$

Construct a distribution with

- most mass on an easy point x_0;
- small, equal masses on the other $d-1$ points;
- a target chosen from all labelings of those points.

If the sample contains few rare points, many labels remain unseen. Because the set is shattered, those labels are unconstrained by observed data.

On an unseen rare point, any algorithm is effectively guessing a fair coin under a random target labeling.

Realizable Lower Bound: Proof Sketch

1. Choose rare-point mass so the sample sees at most about half of them with constant probability.
2. Average over a uniform random labeling of the shattered points.
3. On every unseen rare point, the algorithm's expected error is $1/2$.
4. Use the probabilistic method to select one fixed target attaining the average lower bound.
5. A Chernoff bound supplies the constant-probability event on the number of observed rare points.

The argument is information-theoretic and applies even to computationally unlimited learners.

Lower Bound: Non-Realizable Case

For $\mathrm{VCdim}(\mathcal{H}) = d > 1$, any learning algorithm faces a distribution for which

$$\boxed{\mathbb{P}_{S\sim\mathcal{D}^m}\left[R_{\mathcal{D}}(h_S) - \inf_{h\in\mathcal{H}} R_{\mathcal{D}}(h) > \sqrt{\frac{d}{320m}}\right] \geq \frac{1}{64}.}$$

Hence constant-confidence agnostic learning requires

$$\boxed{m = \Omega\left(\frac{d}{\varepsilon^2}\right).}$$

Intuition: each shattered point carries a small unknown label bias; identifying d weak biases is a coin-estimation problem.

Anthony and Bartlett (1999); Chapter 3 non-realizable lower bound.

What the Three Complexity Measures Tell Us

	Rademacher	**Growth function**	**VC dimension**
Measures	noise fitting	number of dichotomies	largest shattered set
Depends on	sample / distribution	worst m-point set	class only
Applies to	real-valued classes	binary classes	binary classes
Main use	sharp upper bounds	bridge to counting	upper and lower rates
Output	$\widehat{\mathfrak{R}}_S(\mathcal{G})$	$\Pi_{\mathcal{H}}(m)$	integer d

Chapter 3's central chain is

$$\text{VC dimension} \Rightarrow \text{growth bound} \Rightarrow \text{Rademacher bound} \Rightarrow \text{generalization.}$$

Lower bounds show that dependence on d is necessary.

References and Questions

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Questions?

Core mathematical content follows Chapter 3; diagrams, timing, checkpoints, and numerical illustrations are instructor-created.